MCU / 64MHz HSI + HSI48 USB

USB-C + ESD + VBUS detection

TWO PARALLEL POWERED CAN PORTS

12V local branch + reverse polarity + ideal-diode OR + TVS + 3V3

Input examples

Prototype: no Cyphal traffic; transceiver remains in standby.

No onboard 120R termination. Use an end plug at the final module.

Connector pinout is provisional. NOT telephone/Ethernet compatible.

SWD and reset

4P4C powered CAN: qualified straight-through cable, never phone equipment
FABRICATION REVIEW — FOUR-LAYER + MILLING PROTOTYPE

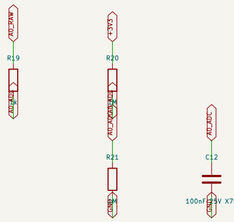
OpenSimGear

Sheet: /
File: starter.kicad_sch

Title: OpenSimGear generic STM32G0B1 module

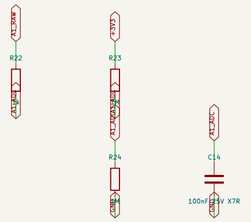
Size: A2	Date: 2026-10-12	**Rev: B draft**
KiCad E.D.A. 10.0.6		Id: 1/2





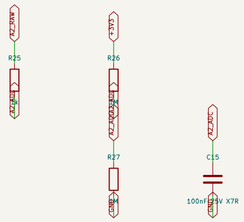
A0: 0-3.3 V input: 1 kΩ / 100 nF filter: 1 MΩ midscale bias





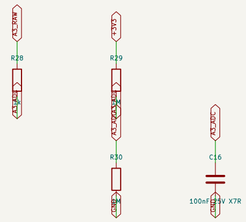
A1: 0-3.3 V input: 1 kΩ / 100 nF filter: 1 MΩ midscale bias



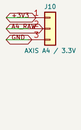


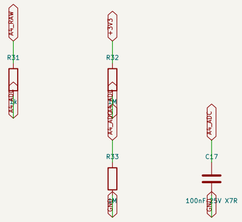
A2: 0-3.3 V input: 1 kΩ / 100 nF filter: 1 MΩ midscale bias





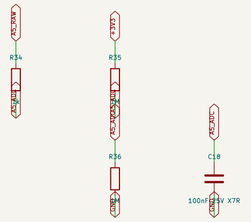
A3: 0-3.3 V input: 1 kΩ / 100 nF filter: 1 MΩ midscale bias





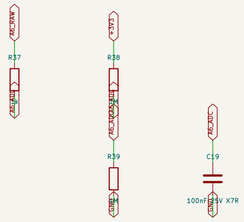
A4: 0-3.3 V input: 1 kΩ / 100 nF filter: 1 MΩ midscale bias





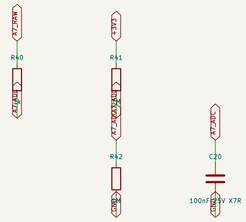
A5: 0-3.3 V input: 1 kΩ / 100 nF filter: 1 MΩ midscale bias





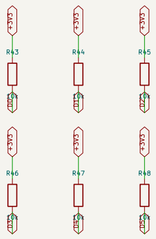
A6: 0-3.3 V input: 1 kΩ / 100 nF filter: 1 MΩ midscale bias





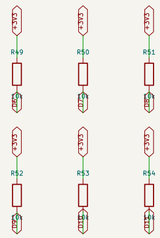
A7: 0-3.3 V input: 1 kΩ / 100 nF filter: 1 MΩ midscale bias





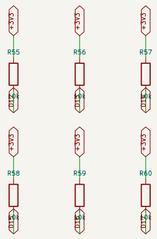
Contact inputs: close signal to GND. No external voltage.





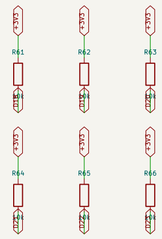
Contact inputs: close signal to GND. No external voltage.





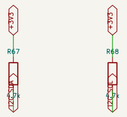
Contact inputs: close signal to GND. No external voltage.





Contact inputs: close signal to GND. No external voltage.





Optional TCA9555 board: 16 active-low inputs per address; keep I2C wiring inside enclosure.
No 5 V / 12 V connection. Pull-ups fitted only on base; validate 400 kHz rise time.

8 analogue axes + 24 active-low digital lines + local 3.3V I2C

Sheet: /Input examples/
File: inputs.kicad_sch

Title: OpenSimGear generic input banks

Size: A2	Date: 2026-10-12	**Rev: B draft**
KiCad E.D.A. 10.0.6		Id: 2/2